

- 19 D** $d \sin \theta = n \lambda$, where $\sin \theta = 1$ at max. order

$$n_{\max} = \frac{d}{\lambda} = \frac{\left(\frac{10^{-3}}{500} \right)}{633 \times 10^{-7}} = 3.16 \rightarrow n = 3$$

number of maximas = 7 (3 maximas on both sides of the central maxima)

- 20 C** Electric field lines point from higher potential to lower potential and must be perpendicular